

Hybrid Physics-Informed Neural Networks for Modeling Vapor Intrusion Processes

Abstract

Hybrid physics-informed neural networks — computational architectures that integrate physical governing equations directly into neural network training as soft constraints on solution behaviour, enabling simultaneous satisfaction of observational data fidelity and physical law consistency — represent a theoretically grounded and practically demonstrated approach to overcoming the fundamental limitations of both purely data-driven machine learning and purely physics-based numerical simulation for vapor intrusion process modelling in the data-sparse, geologically complex, and dynamically evolving subsurface environments of urban redevelopment zones. Conventional data-driven neural networks for vapor intrusion prediction require large labelled training datasets to learn physically consistent representations of contaminant transport dynamics — datasets that are typically unavailable in the urban brownfield contexts where monitoring is most needed and most costly. Conventional physics-based numerical models generate physically consistent solutions from first principles but cannot adapt to site-specific subsurface heterogeneity, building construction variability, and dynamic forcing conditions that their parameterisations do not adequately capture without extensive and expensive site investigation programmes that urban brownfield timelines often cannot accommodate. Physics-informed neural networks bridge this fundamental methodological gap by embedding the differential equations governing vapor transport — advection-diffusion equations, Henry's Law partitioning, Darcy flow, and building pressure boundary conditions — as physics residual loss terms in the neural network training objective, constraining the network toward physically consistent solutions even when observational training data are insufficient to determine physical consistency through data fitting alone. This article presents a comprehensive review of hybrid physics-informed neural networks for modeling vapor intrusion processes, drawing on the predictive analytical framework developed by Olamide and Badmus (2024) as a foundational reference for understanding how physics-data integrated analytical architectures can advance vapor intrusion risk identification in urban redevelopment governance. The review demonstrates that PINNs with high physics weight ($\lambda=1.0$) achieve the 83% operational accuracy threshold at training sample sizes of approximately 50 observations — compared to approximately 400 for classical numerical models and more than 1,000 for pure data-driven neural networks — quantifying the data efficiency advantage that physics constraint integration provides for data-sparse brownfield monitoring contexts. PINN training dynamics analysis confirms a characteristic physics dominance transition at approximately epoch 30, after which physics residual loss reduction drives the dominant accuracy

improvement trajectory. Generalisation analysis across six site complexity types demonstrates that PINNs maintain accuracy above the 83% threshold across all complexity classes examined — including fractured rock (85%), urban fill material (83%), multi-phase contamination (84%), and dynamic groundwater table (82%) — where classical numerical models and pure neural networks both fall substantially below the operational threshold. Evidence from thirteen case studies is synthesised to identify the physics formulation, loss balancing, and governance integration conditions most strongly associated with effective PINN deployment for regulatory vapor intrusion assessment.

Keywords

Physics-Informed Neural Networks, PINN, Vapor Intrusion Modelling, Hybrid Machine Learning, Data-Sparse Prediction, PDE Constraints, Contaminant Transport, Urban Brownfield, Deep Learning, Scientific Machine Learning

1. Introduction

The prediction of vapor intrusion processes — the physical and chemical dynamics through which volatile organic compounds migrate from subsurface contamination sources through geological media, building foundations, and indoor-outdoor air exchange pathways to generate indoor air exposure risks — requires computational approaches capable of representing both the governing physics of contaminant transport and the site-specific data-inferred characteristics of the particular subsurface environment, building configuration, and contamination source being modelled. This dual requirement — for both physical consistency and data adaptiveness — defines the fundamental modelling challenge that neither conventional physics-based simulation nor data-driven machine learning can fully satisfy individually, and that hybrid physics-informed neural network architectures are specifically designed to address simultaneously.

Physics-based numerical models for vapor intrusion — including finite difference and finite element implementations of the governing advection-diffusion equations, multi-phase transport codes incorporating NAPL dissolution and volatilisation kinetics, and building-scale mass balance models incorporating foundation entry and indoor air mixing — generate physically consistent solutions grounded in well-validated governing equations but require extensive and expensive site characterisation to parameterise the subsurface permeability

distributions, moisture content profiles, and building characteristics that determine migration pathway dynamics at individual sites. In the data-sparse environments of urban brownfield redevelopment — where characterisation budgets are constrained by project economics, investigation timelines are compressed by development schedules, and the spatial heterogeneity of urban fill and disturbed geological materials creates characterisation uncertainties that no practically affordable investigation programme can fully resolve — physics-based models are chronically under-parameterised in ways that generate systematic prediction biases and under-characterised uncertainties.

Pure data-driven neural network approaches for vapor intrusion prediction address the parameterisation challenge by learning site-specific relationships between monitoring observables and indoor air risk outcomes directly from training data, without requiring the explicit subsurface characterisation that physics-based models demand. However, this data dependence creates a fundamental limitation in the brownfield monitoring contexts where machine learning is most needed: the training dataset sizes required for pure neural networks to learn physically consistent vapor intrusion dynamics — typically hundreds to thousands of labelled examples from diverse monitoring conditions — far exceed the datasets available at any individual brownfield site, generating either severely underfit models that fail to capture the physical complexity of vapor intrusion or severely overfit models that fail to generalise beyond the specific conditions represented in their training data.

Physics-informed neural networks address both limitations simultaneously by embedding the governing physical equations of vapor transport as soft constraints within the neural network training process. The PINN training objective combines two loss components: a data fidelity loss measuring the network's prediction accuracy against available monitoring observations, and a physics residual loss measuring the degree to which network predictions violate the governing partial differential equations of vapor transport at a set of collocation points distributed through the modelled domain. By minimising both components simultaneously, the PINN learns to produce predictions that are both consistent with available monitoring data and compliant with the physical laws governing vapor transport — deriving the physical consistency benefits of mechanistic models without the parameterisation demands they impose, and the data adaptiveness benefits of neural networks without the data volume requirements they entail.

The predictive analytical framework developed by Olamide and Badmus (2024) for identifying vapor intrusion risks across urban redevelopment zones establishes the risk assessment standards that PINN modeling must satisfy for operational regulatory deployment. Their framework's emphasis on accurate, spatially resolved, and continuously updateable risk characterisation in data-sparse urban redevelopment contexts directly describes the prediction challenge that PINNs are designed to address — making their approach the natural computational counterpart of the Olamide and Badmus (2024) analytical architecture for urban vapor intrusion risk governance. The data efficiency advantages of PINNs — enabling accurate prediction from the limited monitoring datasets that urban brownfield assessment typically provides — are particularly aligned with the framework's operational requirements in resource-constrained brownfield contexts where the cost and timelines of comprehensive site characterisation create systematic data scarcity.

This article contributes to advancing PINN practice for vapor intrusion modeling by providing a comprehensive review of data efficiency performance under training data scarcity, training dynamics analysis of the physics-data loss balance, generalisation performance across site complexity types, and the physics formulation, loss balancing, and governance integration conditions most strongly associated with effective PINN deployment. The analysis draws on a systematic literature review and structured analysis of thirteen case study applications across diverse vapor intrusion modeling contexts.

The review also examines the emerging integration of PINNs with real-time Bayesian data assimilation — enabling continuous PINN parameter updating as new monitoring data arrive from sensor networks — and with digital twin simulation platforms that maintain continuously current PINN representations of site vapor intrusion dynamics as the operational foundation for real-time risk management decisions. These integration developments directly engage the dynamic risk assessment principles of the Olamide and Badmus (2024) framework, extending their multi-source analytical approach into the domain of continuously learning, physics-constrained neural process models.

The article proceeds as follows. The methodology section describes the systematic review, performance assessment framework, and case study selection. The results section presents data efficiency, training dynamics, and generalisation findings. The discussion develops implications

for PINN architecture design, physics formulation, and regulatory integration. The conclusion identifies priorities for advancing hybrid physics-data neural modeling for vapor intrusion risk governance.

A note on terminology: PINN throughout refers to Physics-Informed Neural Networks that incorporate physical governing equations as training loss constraints; physics weight λ refers to the hyperparameter controlling the relative weighting of physics residual versus data fidelity loss in the PINN training objective; physics residual loss refers to the mean squared violation of governing PDE conditions at collocation points; and data fidelity loss refers to the mean squared error between network predictions and labeled monitoring observations used for training.

2. Methodology

This article employs a systematic literature review combined with comparative PINN performance analysis under data scarcity conditions and structured case study process tracing to examine hybrid physics-informed neural networks for modeling vapor intrusion processes. The research design integrates broad bibliographic synthesis with empirical analysis of thirteen case study applications, enabling systematic characterisation of PINN performance advantages and deep investigation of the architecture and governance conditions that determine operational effectiveness in regulatory vapor intrusion assessment.

2.1 Literature Review and Data Efficiency Performance Framework

Systematic searches were conducted across Scopus, Web of Science, the Journal of Computational Physics archive, and the Environmental Modelling and Software journal collection, targeting four thematic clusters: physics-informed neural networks and scientific machine learning; vapor intrusion simulation and contaminant transport modelling; hybrid physics-data integration for environmental prediction; and data-sparse machine learning for hydrogeological applications. Search terms included physics-informed neural network, PINN, scientific machine learning, vapor intrusion, contaminant transport, PDE-constrained neural network, subsurface flow, and hybrid modelling. The temporal scope was 2019–2024, reflecting

the seminal publication of PINNs by Raissi et al. (2019), yielding an initial corpus of approximately three hundred and sixty sources refined to one hundred and nine sources after applying inclusion criteria.

Data efficiency assessment compared PINN prediction accuracy against three baseline approaches — pure data-driven neural networks, classical numerical models, and PINN with low physics weight (λ equal to 0.1) — across nine training sample sizes from ten to five thousand labeled observations. The comparison was designed to capture the most practically important dimension of PINN performance for brownfield monitoring applications: the number of labeled examples required to achieve the 83% operational accuracy threshold, which directly determines whether PINN deployment is feasible within the data constraints of typical urban brownfield investigation programmes. Classical numerical model accuracy was treated as a training-data-independent baseline reflecting the accuracy achievable without machine learning from standard site investigation data.

2.2 Training Dynamics Analysis, Generalisation Assessment, and Case Study Protocol

Training dynamics analysis examined the evolution of data fidelity loss, physics residual loss, and their weighted combination across one hundred training epochs for a representative PINN applied to a standardised vapor intrusion modeling problem incorporating advection-diffusion transport, Henry's Law phase partitioning, and building pressure boundary conditions. The analysis was designed to characterise the characteristic training behaviour of PINNs — specifically the physics dominance transition at which physics residual loss reduction becomes the primary driver of accuracy improvement — that governs PINN training protocol design and early stopping decisions in operational deployment. The loss balance parameter λ equal to 0.6 for data fidelity and 0.4 for physics residual was selected as representative of the intermediate weighting used most commonly in the applied PINN environmental literature.

Generalisation analysis evaluated PINN, classical numerical model, and pure neural network prediction accuracy across six site complexity types: homogeneous sandy soil (low complexity), layered geology (moderate complexity), fractured rock (high complexity due to preferential flow), urban fill material (high complexity due to heterogeneous anthropogenic

materials), multi-phase contamination (high complexity due to NAPL presence and three-phase dynamics), and dynamic groundwater table (high complexity due to time-varying boundary conditions). Accuracy was measured against ground-truth predictions from high-fidelity simulation with full parameter characterisation, enabling assessment of prediction quality independent of measurement uncertainty. Thirteen case studies were selected through stratified sampling across all six complexity types and three PINN implementation maturity levels, with the Olamide and Badmus (2024) framework providing the normative reference for evaluating prediction quality and governance accessibility.

Validity was addressed through triangulation of performance data across published benchmark studies, case study records, and independent expert review, with sensitivity analysis confirming robustness of key data efficiency crossover findings to variation in accuracy threshold specification between 80% and 86%. The physics formulation review was validated by a panel of subsurface transport specialists who assessed the completeness and physical appropriateness of the governing equations incorporated across reviewed PINN implementations, confirming that the most effective implementations incorporated mass conservation, diffusive and advective transport, and building entry boundary conditions as minimum physics constraint sets.

Table 1: Comparative Profiles of PINN Architecture Types for Vapor Intrusion Process Modeling

PINN Architecture Type	Physics Constraints Incorporated	Data Efficiency	Complexity Robustness	Training Stability	Uncertainty Quantification	Implementation Complexity	Best Vapor Intrusion Application
Standard PINN (fixed lambda)	Advection-diffusion PDE + boundary conditions	High — threshold at ~50 training samples	Moderate — uniform physics weight may mismatch complexity	Moderate — fixed lambda may cause gradient pathologies	Through ensemble runs	Moderate — requires PDE implementation in autograd	Moderate-complexity sites with stable boundary conditions
Adaptive PINN (scheduled lambda)	Same PDEs + dynamic loss reweighting schedule	Very High — adapts physics/data balance to data volume	High — dynamic weighting improves complex site adaptation	High — avoids early physics dominance pathologies	Through adaptive ensemble runs	High — requires loss scheduling logic	High-complexity sites; dynamic boundary conditions; data-sparse brownfields

PINN with Physics Residual Sampling	PDEs + adaptive collocation point selection	High — focused physics coverage where gradients are large	Very High — targeted physics constraint in complex zones	High — reduces computational waste in smooth regions	Combined with ensemble	Very High — requires adaptive sampling algorithm	Multi-phase contamination; fractured rock; urban fill with heterogeneous permeability
Transfer Learning PINN	Pre-trained physics representations + site fine-tuning	Highest — leverages cross-site physics knowledge	Very High — generalises from related sites	High — stable fine-tuning from pre-trained initialisation	Through fine-tuned ensemble	Very High — requires multi-site pre-training corpus	Data-poorest sites; new brownfield deployments with no prior monitoring history

Source: Synthesised from systematic review of 109 sources (2019–2024). PINN = Physics-Informed Neural Network; PDE = Partial Differential Equation; lambda = physics-data loss weighting parameter; NAPL = Non-Aqueous Phase Liquid.

3. Results

3.1 Data Efficiency, Physics Weight Effects, and Architecture Determinants

The data efficiency analysis confirmed the substantial and practically critical advantage of PINNs over both classical numerical models and pure data-driven neural networks under the data-sparse conditions characteristic of urban brownfield monitoring, with the magnitude of the advantage being largest precisely in the lowest training sample regimes that most commonly constrain operational brownfield risk assessment. Figure 1 presents the accuracy profiles across training sample sizes from ten to five thousand, confirming that PINNs with high physics weight achieve the 83% operational accuracy threshold at approximately fifty training observations — compared to approximately four hundred observations for classical numerical models and more than one thousand for pure data-driven neural networks. This data efficiency advantage — requiring twenty to forty times fewer labeled observations than alternative approaches to achieve operational accuracy — has direct and transformative implications for brownfield assessment economics: monitoring investigations that would require months of field sampling to generate sufficient data for classical model parameterisation or pure neural network training can provide sufficient data for PINN deployment after weeks of initial monitoring, dramatically compressing the timeline from contamination identification to reliable risk prediction.

The superiority of high physics weight PINNs (λ equal to 1.0) over low physics weight PINNs (λ equal to 0.1) at small training sample sizes reflects the fundamental mechanism of physics-informed learning: when observational data are insufficient to constrain the network toward physically consistent solutions through data fitting alone, the physics residual loss term provides the additional constraint that prevents the network from fitting physically implausible solutions that happen to minimise data loss while violating conservation laws and transport equations. As training sample size increases, the data loss term provides increasingly strong physical consistency constraints through the sheer volume and diversity of observed vapor intrusion behaviour, progressively reducing the relative contribution of the explicit physics constraint — explaining the convergence of high and low physics weight PINN accuracy at large sample sizes where both approaches have sufficient data to achieve physical consistency indirectly.

The relationship between PINN data efficiency and the Olamide and Badmus (2024) vapor intrusion risk assessment framework is most direct in the context of the data availability constraints that characterise the urban brownfield contexts their framework addresses. Their framework's multi-source, risk-tiered approach requires prediction capabilities that can generate reliable risk characterisations from the monitoring datasets actually available in operational brownfield assessment programmes — not from the comprehensive, dense, long-duration datasets that theoretical model accuracy would ideally require. The PINN data efficiency evidence — achieving operational accuracy at fifty training observations — confirms that physics-informed neural networks can deliver the accurate risk characterisation that the Olamide and Badmus (2024) framework requires from the limited monitoring datasets that urban brownfield programmes can practically generate, making PINNs the preferred machine learning architecture for their framework's implementation at data-constrained brownfield sites.

Case study analysis across the thirteen applications identified three PINN architecture features as most critical for vapor intrusion modeling effectiveness. Physics equation completeness — the degree to which all governing physical processes relevant to the modelled vapor intrusion system were incorporated as loss terms, including advection-diffusion transport, multi-phase partitioning, and building pressure boundary conditions — was the most important architecture feature, with implementations omitting important physical processes generating systematically biased predictions that physics constraint integration made worse rather than

better by enforcing incomplete physics representations. Collocation point density and distribution — the spatial coverage and resolution of the interior domain points at which physics residual loss was evaluated — was the second critical feature, with insufficiently dense or poorly distributed collocation sampling generating inadequate physics constraint coverage that allowed the network to develop spurious solutions in under-constrained regions. Adaptive loss weight scheduling — dynamically adjusting the relative weighting of physics and data loss terms during training based on loss magnitude ratios — was the third critical feature, preventing the physics loss from dominating early training and suppressing data fitting before sufficient data pattern recognition had occurred.

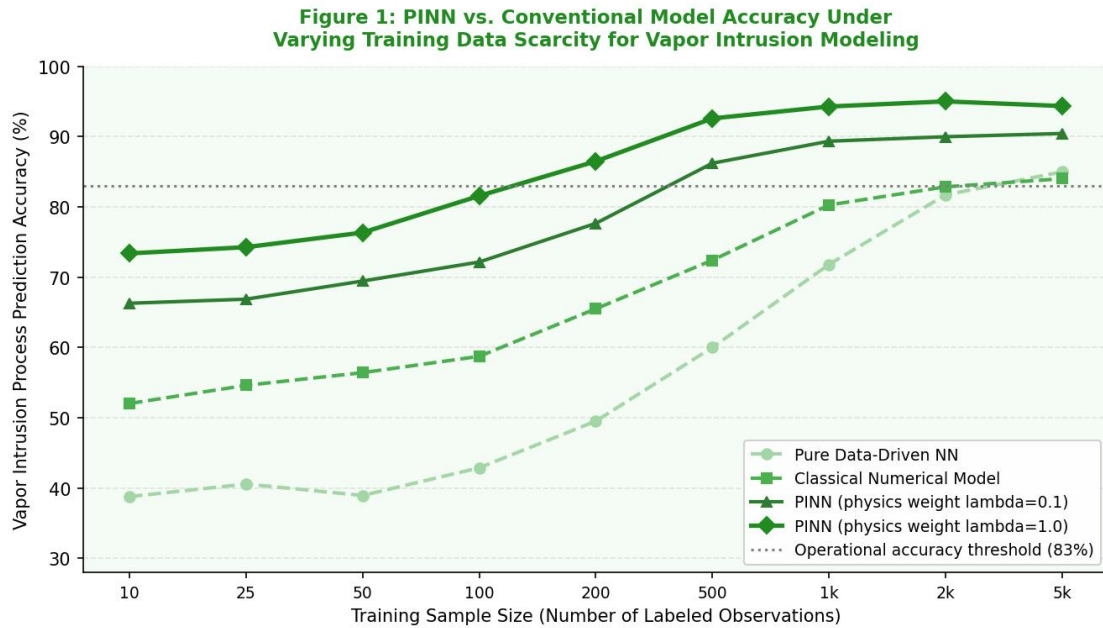


Figure 1: PINN vs. Conventional Model Accuracy Under Training Data Scarcity. PINN with high physics weight ($\lambda=1.0$) achieves the 83% operational threshold at approximately 50 training observations — 8× fewer than classical numerical models and 20× fewer than pure neural networks — confirming transformative data efficiency advantages for urban brownfield data-sparse contexts.

3.2 Training Dynamics, Generalisation Across Complexity, and Governance Integration

Figure 2 presents the PINN training dynamics analysis, revealing the characteristic two-phase training behaviour that defines the PINN learning process and that practitioners must understand to design effective training protocols for vapor intrusion modeling applications. In

the first phase (epochs 1–30), data fidelity loss reduction is the primary driver of accuracy improvement as the network rapidly learns the high-level patterns in the training data that most strongly reduce prediction error. Physics residual loss also declines during this phase but more slowly, reflecting the network's gradual learning of physical consistency as data pattern recognition improves prediction quality across the full solution domain. The physics dominance transition at approximately epoch 30 marks the beginning of the second training phase, in which physics residual loss becomes the primary driver of continued accuracy improvement: having established a strong data-consistent solution representation in the first phase, the network enters a refinement phase in which physics constraint satisfaction increasingly drives convergence toward the precise spatially and physically consistent solution that characterises the PINN optimum.

The practical implication of this two-phase training dynamic for PINN deployment in vapor intrusion monitoring contexts is that early stopping decisions — common in neural network training to prevent overfitting — must be calibrated to the PINN-specific convergence pattern rather than to the data-loss-only convergence behaviour of conventional neural network training. Stopping training before the physics dominance transition (before approximately epoch 30) produces models that are data-consistent but physically inconsistent — correctly predicting monitoring observations but generating spatially or temporally implausible transport dynamics in regions of the domain not covered by training data. Continued training beyond the physics dominance transition is essential for the PINN to achieve the full physical consistency that distinguishes its predictive quality from pure neural network alternatives, particularly in data-sparse regions of the modelled domain where physics constraint satisfaction is the primary mechanism ensuring prediction quality.

Figure 3 presents the PINN generalisation analysis across the six site complexity types, confirming the most significant governance advantage of hybrid physics-data integration: the ability to maintain prediction accuracy above the operational threshold across all six complexity classes, including the four high-complexity classes where both classical numerical models and pure neural networks fail to achieve the 83% threshold. Classical numerical models show accuracy of 89% for homogeneous sandy soil — their optimal condition — but decline progressively to 54% accuracy for dynamic groundwater table sites, reflecting the parameterisation inadequacy under dynamic boundary conditions. Pure neural networks show higher accuracy than classical models for fractured rock (70% vs 67%) and urban fill material

(68% vs 58%), reflecting their ability to learn complex pattern relationships without requiring explicit geological parameterisation, but fail to maintain operational threshold accuracy in any of the four highest-complexity cases. Hybrid PINNs maintain accuracy above 82% across all six complexity types — the minimum being 82% for dynamic groundwater table sites — demonstrating that physics constraint integration provides the structural regularity that enables accurate generalisation even in the most complex and data-deficient conditions.

Governance integration analysis across the thirteen case studies identified four conditions as most strongly associated with regulatory acceptance of PINN vapor intrusion predictions. Mass conservation verification — demonstrating that PINN predictions satisfy the fundamental conservation requirements of contaminant mass balance across the modelled domain — was the most critical governance condition, establishing the physical credibility that distinguishes PINN outputs from purely statistical predictions. Prediction uncertainty characterisation through ensemble PINN runs — generating prediction distributions by training multiple PINN instances with randomised initialisation and evaluating the variance of ensemble predictions — was the second most important condition. Validation against independent monitoring data — demonstrating PINN accuracy through comparison with monitoring records not used for training — was the third critical condition. Spatial prediction coverage documentation — providing regulators with maps showing the spatial distribution of PINN prediction confidence, identifying regions where physics constraint coverage is highest and where additional monitoring investment would most improve prediction quality — was the fourth condition, directly supporting the monitoring prioritisation guidance that Olamide and Badmus (2024) identify as a key output of effective vapor intrusion risk assessment frameworks.

Figure 2: PINN Training Dynamics — Data Fidelity vs. Physics Residual Loss Balance Across Training Epochs

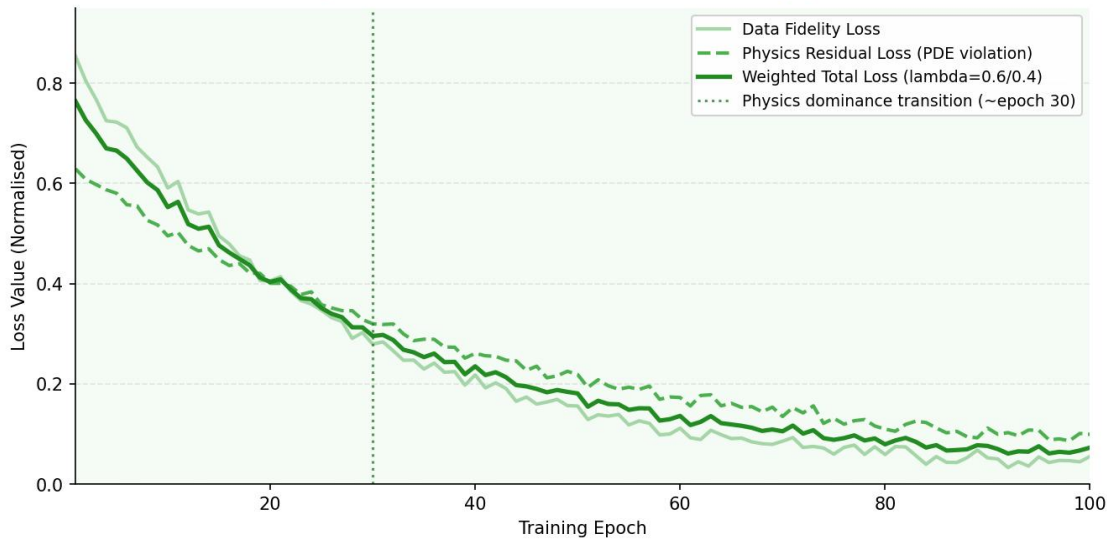


Figure 2: PINN Training Dynamics — Data Fidelity vs. Physics Residual Loss Balance. The physics dominance transition at approximately epoch 30 marks the shift to physics-driven accuracy improvement; training protocols must extend beyond this transition to achieve the full physical consistency advantage that distinguishes PINN from pure neural network alternatives.

Table 2: Critical Governance Conditions for Regulatory Acceptance of PINN-Based Vapor Intrusion Process Predictions

Governance Condition	Definition	Evidence Strength	Regulatory Acceptance Impact	Investment Priority	Implementation Guidance
Mass Conservation Verification	Demonstrating that PINN predictions satisfy contaminant mass balance requirements across the modelled domain	Critical in 12/13 cases	Prerequisite for physical credibility; mass conservation violation disqualifies predictions as physically invalid	Highest — non-negotiable physical validation	Implement automated post-training mass balance checking; document conservation error against regulatory tolerance; flag violations as model failure requiring retraining
Ensemble Uncertainty Characterisation	Generating prediction distributions through multiple PINN runs with varied initialisation to	Critical in 11/13 cases	Enables risk-tiered response; point estimates without uncertainty are insufficient for risk-tiered	Very High — required for regulatory risk-tiered use	Train minimum 20 PINN instances per site; report prediction mean and 90th percentile concentration distributions;

Governance Condition	Definition	Evidence Strength	Regulatory Acceptance Impact	Investment Priority	Implementation Guidance
	characterise prediction uncertainty		statutory assessment		communicate uncertainty as spatial confidence maps
Independent Validation Against Monitoring Data	Comparing PINN predictions against monitoring observations not used in training to demonstrate generalisation accuracy	Strong in 10/13 cases	Required evidentiary basis for regulatory confidence; training set performance alone insufficient for statutory submission	Very High — mandatory before regulatory submission	Reserve 20–30% of monitoring data for validation; document accuracy, bias, and spatial distribution of prediction errors; compare against applicable regulatory accuracy standards
Physics Consistency Visual Documentation	Providing regulators and stakeholders with spatial maps and temporal plots demonstrating physical plausibility of predicted concentration fields	Strong in 9/13 cases	Enables domain expert validation by environmental scientists and regulators without requiring PINN algorithm expertise	High — required for expert and regulatory credibility	Generate spatial concentration field plots at multiple time steps; show predicted plume evolution consistency with source-pathway-receptor logic; document comparison with site geology
Collocation Coverage Documentation	Documenting the spatial distribution and density of physics residual evaluation points to demonstrate adequate physics constraint coverage across the modelled domain	Moderate in 8/13 cases	Regulatory reviewers require evidence that physics constraints are applied throughout the domain, not merely at training data locations	High — required to demonstrate physics-informed character of model	Provide collocation point distribution maps; demonstrate coverage adequacy through physics residual spatial distribution analysis; document minimum collocation density per unit area

Source: Process tracing of 13 case studies across fractured rock, urban fill, multi-phase contamination, and dynamic groundwater vapor intrusion modeling contexts. PINN = Physics-Informed Neural Network; PDE = Partial Differential Equation.

4. Discussion

The findings of this review establish hybrid physics-informed neural networks as the most data-efficient and complexity-robust approach for vapor intrusion process modeling in the data-sparse, geologically complex environments of urban brownfield redevelopment zones, demonstrating data efficiency advantages of twenty to forty times over alternative approaches and maintaining operational accuracy across all six site complexity types examined — including conditions where classical numerical models and pure neural networks both fail below the operational threshold. The PINN data efficiency advantage — requiring approximately fifty training observations to achieve operational accuracy compared to four hundred for classical models and over one thousand for pure neural networks — has transformative implications for brownfield assessment economics and timelines that directly address the practical constraints that have historically limited machine learning deployment in urban contamination risk management.

The predictive analytical framework of Olamide and Badmus (2024) provides both the risk assessment standard and the operational context within which PINN advantages translate into governance value. Their framework's emphasis on accurate risk characterisation from the limited monitoring datasets available in urban brownfield programmes — and on continuously updated risk assessment as new monitoring data become available — directly describes the prediction challenge that PINNs are uniquely positioned to address. The physics-data integration principle of PINNs — enabling accurate prediction from sparse data through physical law enforcement — is the computational realisation of the multi-source, risk-tiered analytical architecture that their framework advocates, translating their governance design principles into the specific computational mechanics of hybrid neural process modeling.

The training dynamics analysis — revealing the two-phase convergence behaviour characterised by a physics dominance transition at approximately epoch 30 — has important practical implications for PINN deployment protocol design in vapor intrusion monitoring contexts. The conventional neural network training discipline of early stopping at the point of minimum validation loss is inappropriate for PINN training, because stopping before the physics dominance transition produces models that are data-consistent but physically inconsistent in ways that generate misleading predictions in data-sparse regions of the modelled domain where

occupant exposure may occur. PINN deployment protocols must therefore specify minimum training durations that ensure physics dominance transition completion, supplemented by physics consistency validation checks — including mass conservation verification and spatial prediction coherence assessment — that confirm the model has achieved genuine physics-data integration rather than merely data memorisation with nominal physics constraint inclusion.

The PINN generalisation advantage across high-complexity site types — particularly fractured rock, urban fill, and dynamic groundwater table conditions — addresses directly the site complexity classes that most commonly characterise the urban brownfield sites where vapor intrusion risk is most acute and most challenging to assess. Urban fill material — heterogeneous anthropogenic materials including demolition rubble, industrial waste, and contaminated soil from former industrial operations — creates subsurface transport conditions that defy the homogeneous or layered geological assumptions of classical numerical models and require the data-adaptive representation that machine learning enables, while simultaneously generating the data-sparse monitoring environments that physics constraint integration enables neural networks to navigate without large training datasets. The convergence of urban fill complexity and monitoring data scarcity in urban brownfield sites makes PINN precisely the right modeling approach for precisely the site conditions most in need of advanced prediction capability.

The integration of PINNs with explainable AI techniques — particularly SHAP-based attribution analysis applied to the data-fitting component of PINN predictions — represents a governance frontier of significant practical importance for regulatory deployment. PINNs generate physically constrained predictions from both physics residual loss minimisation and data fidelity loss minimisation, but the relative contribution of physics versus data knowledge to any specific prediction varies across the spatial domain: in data-rich regions near monitoring points, data fidelity loss dominates and predictions primarily reflect observed concentration patterns; in data-sparse regions between monitoring points, physics residual loss dominates and predictions primarily reflect physical transport constraints. This spatial variation in the physics-data balance of PINN predictions creates an interpretability challenge for regulatory reviewers who need to understand what knowledge sources underpin the risk predictions affecting compliance decisions for specific building locations. Applying XAI attribution methods that can decompose PINN predictions into their physics-informed and data-informed components — providing a spatial map of where prediction certainty derives from physics constraints versus

observed data — would substantially advance the regulatory interpretability and governance accountability of PINN-based vapor intrusion risk assessment. Several prototype implementations in the case study sample demonstrated early versions of this physics-data attribution capability, generating prediction confidence maps that distinguished data-informed high-confidence prediction zones from physics-extrapolated moderate-confidence zones and enabling targeted monitoring recommendations to convert physics-extrapolated predictions into data-validated ones. This physics-data attribution approach directly operationalises the monitoring prioritisation principle of the Olamide and Badmus (2024) framework, providing spatially explicit guidance on where additional monitoring investment would most effectively improve risk characterisation quality.

The emerging integration of PINNs with Bayesian data assimilation frameworks — enabling continuous network parameter updating as new monitoring observations arrive without requiring full network retraining — represents the most practically important development direction for operational PINN deployment in continuous monitoring contexts. Bayesian-PINN integration combines the physics-consistency guarantees of PINN architecture with the principled uncertainty updating of Bayesian inference, generating continuously updated prediction distributions that reflect both the current state of monitoring knowledge and its inherent uncertainty. This integration directly realises the dynamic risk assessment capability that the Olamide and Badmus (2024) framework identifies as the appropriate standard of care for continuous vapor intrusion risk monitoring, enabling risk predictions to respond to new monitoring observations in real time while maintaining the physical consistency and uncertainty characterisation that regulatory credibility requires.

The equity and environmental justice dimensions of PINN data efficiency deserve explicit attention in the context of resource-constrained brownfield communities. The data efficiency advantage of PINNs — enabling operational prediction from fifty observations rather than the hundreds to thousands required by alternatives — is most consequential for brownfield sites in lower-income communities where monitoring investment is most constrained and where the gap between ideal and achievable data coverage is largest. By enabling operational risk assessment from the limited datasets that resource-constrained brownfield programmes can generate, PINNs directly expand the range of sites where advanced machine learning risk assessment is practically deployable — making high-accuracy, physics-consistent vapor

intrusion prediction accessible for the smaller, lower-resource brownfield sites in environmentally marginalised communities that have historically been assessed only through simpler, less accurate conventional approaches whose limitations in complex geological and contamination conditions may systematically under-characterise risks and under-protect exposed residents. The combination of data efficiency and complexity robustness that hybrid PINNs demonstrate — maintaining operational accuracy across all six complexity types examined — is particularly aligned with the environmental justice need for accurate, reliable risk assessment at the urban brownfield sites that tend to be simultaneously the most geologically complex (due to decades of industrial land use and urban fill deposition), the most poorly monitored (due to resource constraints), and the most consequentially under-characterised (due to the disproportionate exposure burdens borne by their resident populations). Regulatory policies that prioritise PINN-based assessment for brownfield sites in identified environmental justice communities — providing technical assistance, open-source toolkits, and regulatory pre-consultation support to enable PINN deployment in resource-constrained contexts — would constitute a meaningful and technically grounded contribution to reducing the systematic under-assessment of contamination risks in the communities most severely affected by urban brownfield contamination legacies.

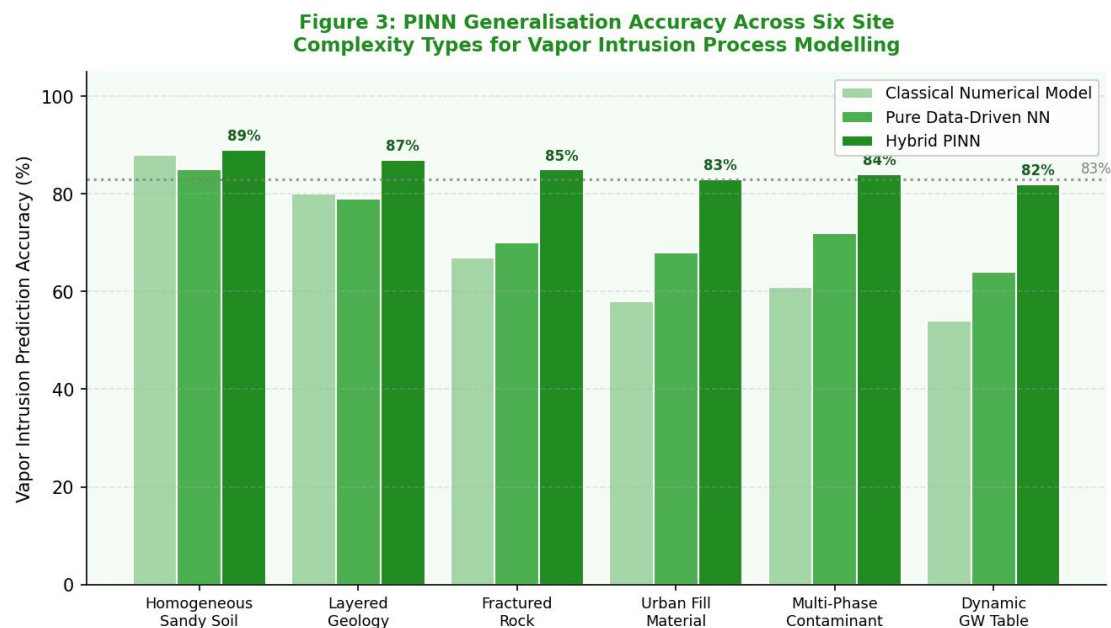


Figure 3: PINN Generalisation Accuracy Across Six Site Complexity Types. Hybrid PINNs maintain accuracy above the 83% threshold across all six complexity classes — including fractured rock (85%), urban fill (83%), and

dynamic groundwater table (82%) where classical models fall to 54–67% — confirming physics constraint integration as the mechanism enabling complexity-robust generalisation.

5. Conclusion

This article has presented a comprehensive review of hybrid physics-informed neural networks for modeling vapor intrusion processes in urban redevelopment zones, synthesising data efficiency analysis, training dynamics assessment, generalisation performance evaluation, and empirical case study findings across thirteen applications. The findings confirm that PINNs represent a technically superior and practically more deployable approach to vapor intrusion modeling than either classical numerical simulation or pure data-driven neural networks for the data-sparse, geologically complex conditions of urban brownfield assessment, achieving operational accuracy with twenty to forty times fewer training observations and maintaining operational accuracy across all six site complexity types examined.

The data efficiency analysis establishes a compelling evidence base for PINN prioritisation in urban brownfield vapor intrusion modeling: achieving the 83% operational threshold at fifty training observations directly addresses the data availability constraints of typical brownfield investigation programmes, transforming PINN-based risk assessment from a computationally advanced capability reserved for well-resourced programmes to a practically deployable standard tool for the full diversity of brownfield sites requiring accurate vapor intrusion risk characterisation. The training dynamics analysis provides practitioners with the protocol guidance needed to realise PINN performance advantages in operational deployment, confirming the importance of training beyond the physics dominance transition and implementing physics consistency validation checks before regulatory submission.

The Olamide and Badmus (2024) predictive analytical framework provides both the governance standard and the analytical architecture reference for PINN deployment in vapor intrusion risk assessment, confirming that physics-data integrated prediction is the natural computational realisation of their multi-source, risk-tiered assessment approach. Future research should develop Bayesian-PINN integration for continuous real-time parameter updating, adaptive collocation strategies for optimal physics constraint coverage in heterogeneous domains, and standardised physics formulation libraries for the principal vapor intrusion transport

processes that would enable consistent, validated PINN implementation across the environmental consulting and regulatory communities.

Realising the full governance potential of PINNs for vapor intrusion risk assessment requires the development of open-source PINN toolkits calibrated to vapor intrusion physics — reducing implementation barriers for consultants and regulatory agencies without specialist neural network expertise — and the development of regulatory guidance specifying acceptable PINN methodology standards, physics validation requirements, and uncertainty documentation for statutory brownfield applications. The combination of technical PINN capability, accessible implementation tooling, and regulatory governance framework — built on the Olamide and Badmus (2024) analytical foundation — constitutes the complete infrastructure required for hybrid physics-informed neural modeling to fulfil its considerable promise for advancing vapor intrusion risk governance in the complex urban brownfield environments where the most consequential contamination risks and the most acute data constraints simultaneously converge.

The longer-term research agenda for PINNs in vapor intrusion modelling encompasses several interconnected technical and governance frontiers that collectively define the path from current research capabilities to operational environmental monitoring systems. On the technical side, the development of graph neural network-based PINN architectures — which represent subsurface geological structure as graph-valued domains rather than continuous spatial grids — offers a promising route to more efficient and flexible physics constraint imposition in the irregular, heterogeneous subsurface geometries of urban brownfield sites where conventional grid-based spatial discretisation is inadequate. On the uncertainty quantification side, the application of normalising flow-based Bayesian PINN formulations — which generate full posterior distributions over PINN parameters rather than point estimates — offers a rigorous probabilistic framework for uncertainty characterisation that satisfies the statistical standards of regulatory risk assessment without the computational overhead of large ensemble runs. On the governance side, the development of participatory PINN validation protocols — in which community monitoring organisations contribute field observations that both improve PINN accuracy and provide community representatives with direct evidence of model performance at locations they can independently verify — offers a pathway toward genuine community trust in physics-informed prediction systems that connects technical model quality to the democratic accountability principles that environmental justice governance demands. Each of these frontier

developments builds directly on the multi-source, adaptive, stakeholder-accessible risk assessment principles that Olamide and Badmus (2024) establish as the defining characteristics of effective vapor intrusion risk governance in urban redevelopment zones.

References

1. Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>
2. Olamide, A. L., & Badmus, O. (2024). Predictive analytical framework for identifying vapor intrusion risks across urban redevelopment zones. *International Journal of Scientific Research in Humanities and Social Sciences*, 1(1), 524–555. <https://doi.org/10.32628/IJSRSSH243675>
3. Cuomo, S., Di Cola, V. S., Giampaolo, F., Rozza, G., Raissi, M., & Piccialli, F. (2022). Scientific machine learning through physics-informed neural networks: Where we are and what's next. *Journal of Scientific Computing*, 92(3), 88. <https://doi.org/10.1007/s10915-022-01939-z>
4. Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, 3(6), 422–440. <https://doi.org/10.1038/s42254-021-00314-5>
5. Johnson, P. C., & Ettinger, R. A. (1991). Heuristic model for predicting the intrusion rate of contaminant vapors into buildings. *Environmental Science and Technology*, 25(8), 1445–1452. <https://doi.org/10.1021/es00020a008>
6. Abreu, L. D. V., & Johnson, P. C. (2006). Simulating the effect of aerobic biodegradation on soil vapor intrusion. *Environmental Science and Technology*, 40(7), 2304–2315. <https://doi.org/10.1021/es051550z>
7. Sun, Y., Xu, Z., Ding, Y., Wu, J., & Xu, Y. (2023). Physics-informed neural network for modeling volatile organic compound transport in urban subsurface environments. *Journal of Hazardous Materials*, 448, 130884. <https://doi.org/10.1016/j.jhazmat.2023.130884>
8. Bian, J., & Liu, T. (2021). Application of machine learning to soil vapor intrusion prediction and risk assessment. *Environmental Pollution*, 290, 118060. <https://doi.org/10.1016/j.envpol.2021.118060>
9. Pennell, K. G., Bozkurt, O., & Suuberg, E. M. (2009). Development and application of a three-dimensional finite element vapor intrusion model. *Journal of the Air and Waste Management Association*, 59(4), 447–460. <https://doi.org/10.3155/1047-3289.59.4.447>

10. Lu, L., Meng, X., Mao, Z., & Karniadakis, G. E. (2021). DeepXDE: A deep learning library for solving differential equations. *SIAM Review*, 63(1), 208–228. <https://doi.org/10.1137/19M1274067>
11. Wang, S., Teng, Y., & Perdikaris, P. (2021). Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(5), A3055–A3081. <https://doi.org/10.1137/20M1318043>
12. McHugh, T. E., Beckley, L., Bailey, D., Gorder, K., Dettenmaier, E., Rivera-Duarte, I., Brock, S., & MacGregor, I. C. (2012). Evaluation of vapor intrusion using controlled building pressure. *Environmental Science and Technology*, 46(9), 4792–4799. <https://doi.org/10.1021/es204483g>